

Comprehensive KPI & Performance Metrics Compendium

Quantitative Evaluation across Cohort ($N = 41,117$), AI Model VM, Edge TinyML & 5G

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Executive KPI Overview: This document aggregates every quantitative Key Performance Indicator (KPI), benchmark metric, mathematical formula, statistical process control alarm, and latency budget calculated across the master’s research report (Pamit_Duggal_Internship_Report_Final.pdf) and validated on real-world industrial data from Linde HomeCare France.

1 National Cohort Clinical & Adherence KPIs ($N = 41,117$)

The clinical dataset provided by **Linde HomeCare France** contains anonymized telemonitoring records representing 41,117 patients undergoing CPAP therapy for Obstructive Sleep Apnea (OSA).

Clinical Metric / Indicator	Value / Mean	Count	Clinical Threshold & Regulatory Context
Total Validated Cohort Size	41,117	100.0%	National telemonitoring cohort across France
7-Day Mean Nightly Usage Duration	6.71 h/night	Std: 2.15 h	High baseline compliance in active population
Statutory Compliance (CPAM Rule)	90.12%	37,056 pts	≥ 4.0 h/night on $\geq 70\%$ of nights (Reimbursement line)
Sub-Therapeutic Usage	8.07%	3,318 pts	< 4.0 h/night → Targeted for automated video coaching
Critical Non-Adherence / Zero Usage	1.81%	743 pts	0.0 h/night → Urgent technician outreach required
7-Day Mean Residual AHI	3.07 /hr	Median: 1.83	Normal therapeutic apnea suppression (< 5.0 /hr)
Fully Controlled Therapy	81.75%	33,615 pts	Normal breathing restored ($AHI < 5.0$ /hr)
Mild Residual Apnea	14.28%	5,871 pts	$5.0 \leq AHI < 15.0$ /hr
Moderate Residual Apnea	3.62%	1,488 pts	$15.0 \leq AHI < 30.0$ /hr
Severe Residual Apnea Spike	0.35%	143 pts	$AHI \geq 30.0$ /hr → Immediate physician escalation
95th Percentile Mask Leakage Mean	18.42 L/min	Std: 12.65	Normal baseline seal (< 24.0 L/min)
Significant Mask Leak Episodes	14.85%	6,106 pts	Leak ≥ 24.0 L/min for > 15 continuous minutes

Table 1: National Cohort Clinical Performance and Statutory Adherence Indicators ($N = 41,117$)

2 AI Model VM Pipeline KPIs (Port 8000)

The AI Model VM runs my supervisor’s 7-layer predictive pipeline (M4_FINAL_F2.ipynb) wrapped inside api_data_loader.py, calculating statistical process control drift, patient phenotyping, and survival risks.

2.1 Layer 0: Statistical Process Control (SPC) Surveillance Alarms

- **Total Active Surveillance Alarms Triggered: 12,112 alarms** (29.46% of the cohort flagged).
- **CUSUM Usage Degradation Alarms: 5,938 alarms** (Detects persistent cumulative drops below 56-night baseline).
- **CUSUM AHI Spike Alarms: 7,588 alarms** (Detects acute spikes in residual apnea events).
- **EWMA Trend Alarms ($\lambda = 0.2$): 2,628 alarms** (Exponentially weighted multi-day downward drift).

2.2 Layers 1 to 3: Risk Stratification, Phenotyping & Survival Hazards

Risk Tier / Feature	Proportion	Patient Count	Automated System Action Dispatched
Mean Cohort Risk Metric (z_{risk})	0.1977	41,117 pts	Composite multi-specialist risk score $\in [0, 1]$
Low Risk Tier ($z_{\text{risk}} < 0.3$)	76.58%	31,487 pts	Standard passive telemonitoring; no intervention needed
Medium Risk Tier ($0.3 \leq z_{\text{risk}} < 0.7$)	23.40%	9,621 pts	Dispatched automated personalized video coaching
High Risk Tier ($z_{\text{risk}} \geq 0.7$)	0.02%	9 pts	Triggered urgent physical technician home visit

Table 2: 3-Tier Risk Stratification and Action Dispatch Distribution ($N = 41,117$)

2.3 Explainable AI (SHAP) Feature Attribution Ranking

Using model-agnostic **SHapley Additive exPlanations (SHAP)**, the top feature drivers predicting 90-day dropout are:

- 95th Percentile Mask Leakage (SHAP = +0.42):** Strongest single contributor to early therapy abandonment.
- Early Titration Usage Slope (Days 1–14) (SHAP = +0.38):** Negative usage slope in the first 2 weeks strongly forecasts permanent dropout.
- Residual AHI Variability (SHAP = +0.29):** Micro-arousals and oxygen fluctuations preventing restorative sleep.
- Care Intensity / Pending Visit Age (SHAP = +0.18):** Delayed clinician response following reported side-effects.

2.4 Layer 5: Multi-Sensor Connected Biomarker Anomalies ($N = 357$ active patients)

- Cardiac Arrhythmia / Atrial Fibrillation: 27 patients** flagged via Withings BPM Core digital ECG.
- Nocturnal Hypoxia ($\text{SpO}_2 < 88\%$): 103 patients** flagged via Masimo Rad-G pulse oximetry.
- Morning Severe Hypertension (≥ 140 mmHg): 14 patients** flagged via Withings BPM Core.
- REM Sleep Fragmentation / Arousals: 68 patients** flagged via SomnoArt digital sleep staging.

3 Edge Computing & TinyML Optimization KPIs (Raspberry Pi 5)

Engineering Parameter	Baseline (FP32)	Quantized (INT8)	Optimization Gain / Clinical Impact
Model Memory Footprint	2.40 MB	486.2 kB	80.0% RAM Reduction
Inference Latency (ARM Cortex-A76)	845.2 ms	263.9 – 313.5 ms	63.8% Faster ; enables sub-second triage
Mean Absolute Error (MAE) Drift	Ref (0.0)	< 0.0042 L/min	Zero clinically significant precision loss
Single-Core CPU Utilization	78.4%	< 12.0%	Leaves > 88% CPU free for background ETL
Sensor Window Duration	20.0 seconds	20.0 seconds	Standard respiratory window analyzed in < 0.32s

Table 3: Edge TinyML Quantization and Latency Benchmarks on Raspberry Pi 5

3.1 Physical Edge Device Real-Time Execution Logs

Event ID	Detected Anomaly Trigger	Severity	Inference	Transit	Total Edge	SLA Status
4ec7b5d7...	Mask Style Change / High Leak	High	263.9 ms	7.4 ms	271.3 ms	Passed ($\leq 60s$)
73ab6166...	Nasal Congestion / Dry Mouth	Low	313.5 ms	45.9 ms	359.4 ms	Passed ($\leq 60s$)
ba3067be...	High Leak Titration Threshold	Medium	272.0 ms	144.2 ms	416.2 ms	Passed ($\leq 60s$)
c6951da3...	Nasal Resistance / Pressure	Low	264.0 ms	64.9 ms	328.9 ms	Passed ($\leq 60s$)

Table 4: Real-Time Edge Detection and Wi-Fi Delivery Timings on Raspberry Pi 5

4 Video VM Microservice Performance KPIs (Port 8080)

Microservice Benchmark Metric	Measured Value	Architectural Performance Standard
Curated Clinical Video Catalog	27 / 27 (100.0%)	Full HD (1080p) @ 30.0 FPS across 5 clinical topics
Bilingual WebVTT Subtitle Coverage	54 / 54 (100.0%)	27 English (.en.vtt) and 27 French (.fr.vtt) tracks
Total Video Storage Footprint	241.41 MB	Avoided doubling to 500 MB by decoupling WebVTT
Average Video Module Duration	10.4 seconds	Short, targeted, non-fatiguing coaching clip
Server Health Check Latency (GET /health)	5.75 ms (Min: 2.85 ms)	Sub-10ms non-blocking asynchronous response
Catalog Query Latency (GET /api/library)	13.35 ms	Ultra-fast metadata retrieval
Subtitle Fetch Latency (GET /subtitles/*)	14.22 ms	Instant track mounting on client HTML5 player
Protected Orchestration (POST /orchestrate)	1,090.42 ms	Includes decision verification, logging, and retry
Malicious Crawler Interception	100.0% blocked	Instant HTTP 403 / 429 response via SlowAPI limiter
Unauthorized Token Request Filter	100.0% blocked	Strict rejection of requests missing valid X-API-KEY

Table 5: FastAPI Video Server Performance and Security Metrics (CPAP_VIDEO_SERVER_KPIS.txt)

5 End-to-End Latency Budget & 5G CAMARA QoD KPIs

To satisfy our clinical requirement, video coaching must arrive on the patient’s bedside screen in **under 60.0 seconds** upon waking.

Pipeline Segment	Fast Path (Edge)	Slow Path (Cloud)	Component Description	Description
1. Sensor Ingestion & Feature Extraction	18.5 ms	24.2 ms	Parsing CSV buffer and computing 20s features	
2. INT8 TinyML Anomaly Classification	285.0 ms	285.0 ms	Local inference on ARM Cortex-A76	
3. AI Model VM 7-Layer Risk Scoring	<i>Bypassed</i>	450.0 ms	CUSUM, PCA phenotyping, and survival hazard	
4. Video Decision & DB Logging	8.2 ms	12.8 ms	Catalog lookup and SQL audit record write	
5. 5G CAMARA QoD Network Slicing	<i>Local Wi-Fi</i>	75.4 ms	Guaranteed radio bearer slice request	
6. WebVTT Subtitle Mounting & Player Init	14.2 ms	14.2 ms	Attaching synchronized French/English track	
7. Dashboard UI Rendering	25.0 ms	25.0 ms	HTML5 DOM video tag mounting	
Total End-to-End Latency	≈ 0.35 seconds	≈ 1.40 seconds	Comfortably beats the 60.0s Morning Budget	

Table 6: End-to-End Latency Budget Breakdown: Fast Path vs. Slow Path

Summary SLA Compliance Rate: 100.0%

Every recorded execution in both local edge testing and Berlin 6G-Path cloud trial resolved in < 2.1 seconds, leaving over 57.9 seconds of safety headroom against the 60-second medical delivery SLA.